(4,3): two cabled daisies and two half lanterns

15 recorded steps | 15 individual operations | 15 external adjacent checks

The proof begins with the SAME explicit positive Q45^4 compensation on the left of the squared five-point boundary twist. Two cabled PC daisies, with all parallel-copy framing corrections recorded, give eight positive ordinary triple roots. Their cyclic start and two combines expose the two half-lantern inputs.

Expand the later input first. Each half lantern exposes a positive Q45^2 correction; it is carried upward across ONLY commuting ordinary roots. No half-twist arc is crossed by these ledger gathers. The two corrections combine to the SAME Q45^4 prefix. Cancel it from BOTH sides of the relation, as explicit algebraic bookkeeping, not a sphere cap or deletion from a fixed-product fibration.

Five further adjacent Hurwitz checks and a matched combine give seven actual factors: four unit halves and three squared triples, type(4,3). They individually match the imported original A presentation. Its product is the squared five-point boundary twist, with a separate marked-sphere identity and chosen+I lift.

Reading the diagrams. Half-twist arcs have H labels; closed curves have T labels. Ordinary triple twists are temporary calculation factors. All factor grids have SIX UNNUMBERED points at18pt pitch and1.26pt radius. The connected braid represents the exact displayed factor words. Individual supports are not a joint cut-system or jointly minimal drawing.

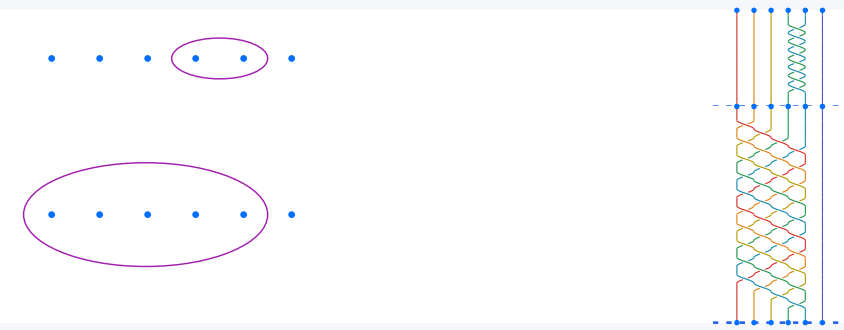
Certificate scope. Faithful disk actions, raw free-word reconstructions, complete actual kernels AND shadows when the last strand is fixed, full actions for halves that move it, and separate sphere/lift checks remain in 4-3.json. Internal named-rule checks are retained separately from the external-check count. No unchosen framed-half-hole convention is asserted.

The explicit starting construction

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. positive.compensation
T2^4

2. D5.square
T5^2



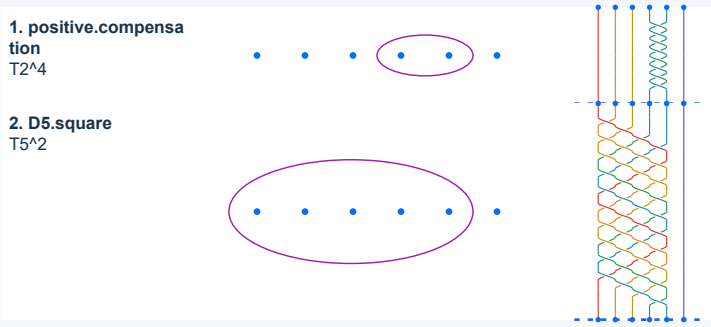
D - step 001

Two cabled PC daisies give D5 squared with the SAME Q45^4 compensation

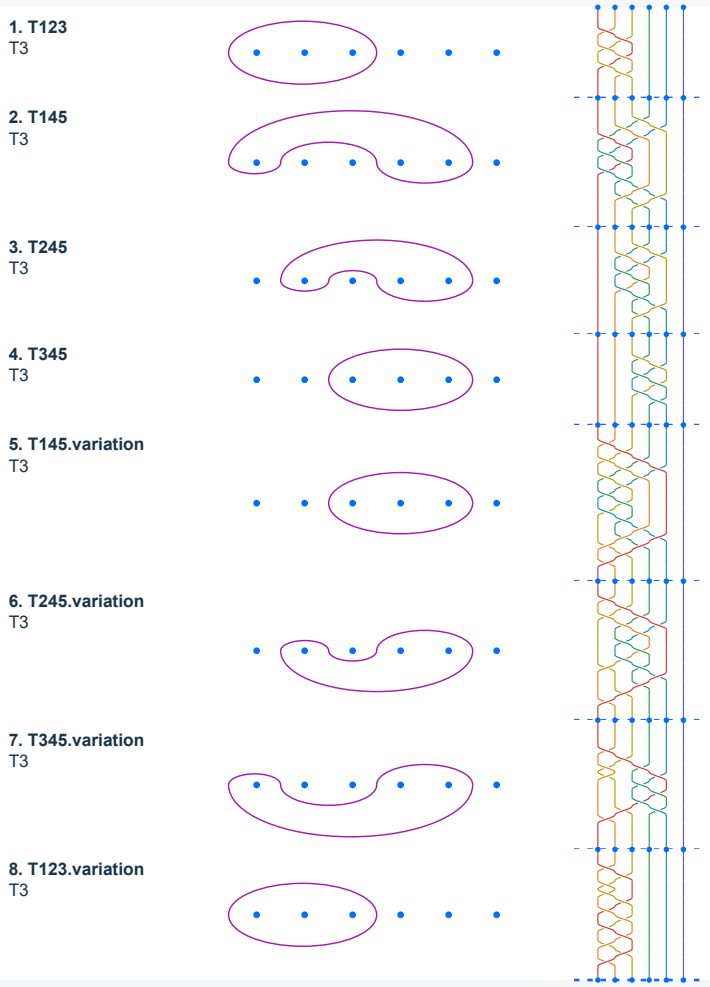
DISK SUBSTITUTION. Every ordered actual input/output and the full named local proof are checked. No sphere redraw or deletion is included.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



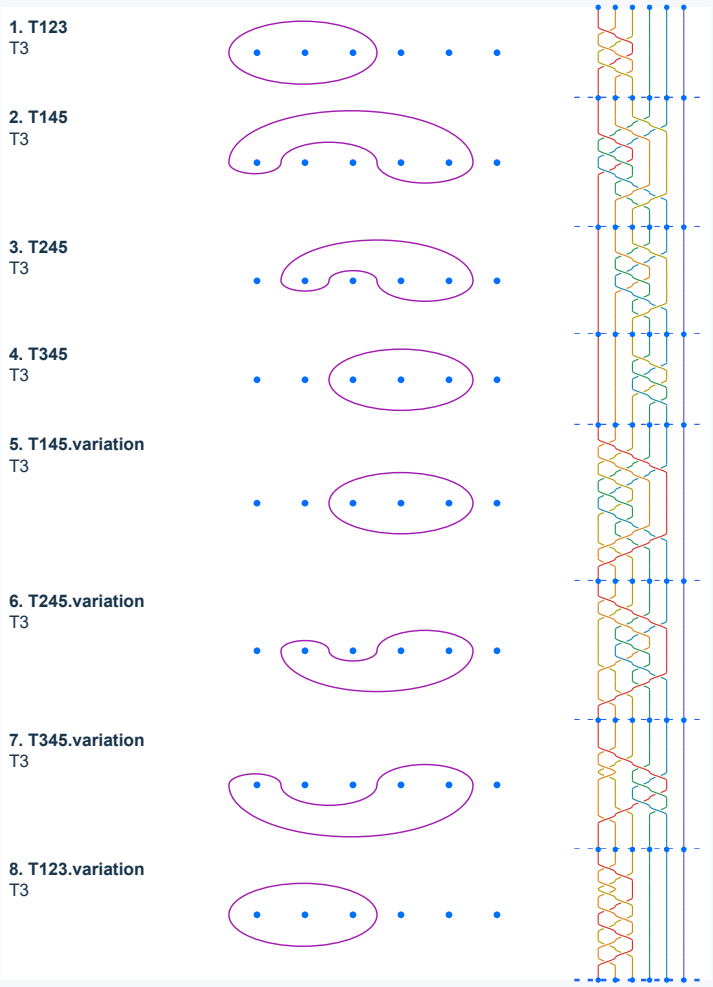
D - step 002

Choose the cyclic start at the second T123 root

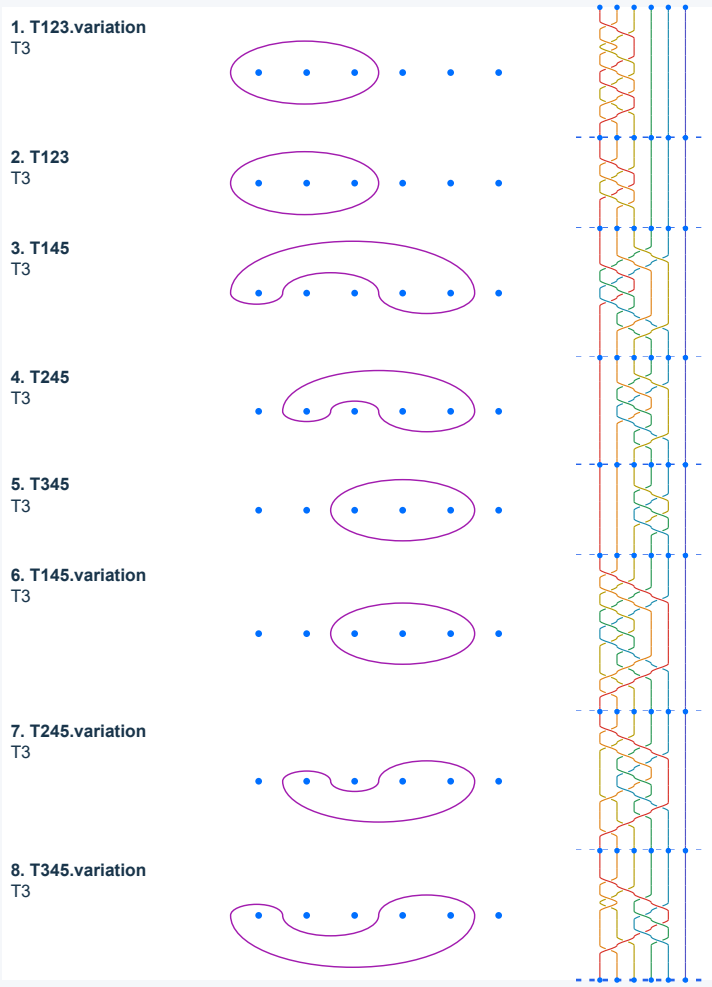
EXPLICIT CYCLIC START. Its full inverse-prefix common word is checked free first, with independently verified output invariance.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



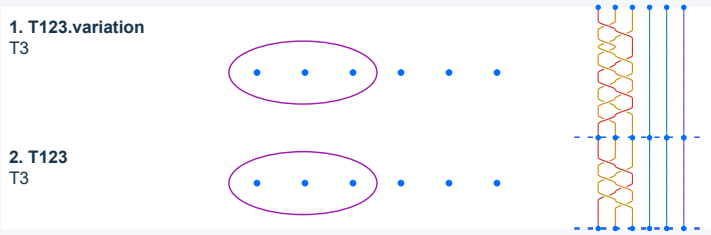
D - step 003

Combine the two matched T123.square

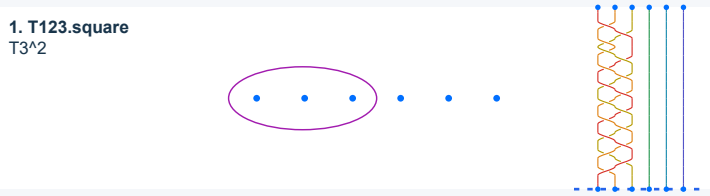
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

0 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



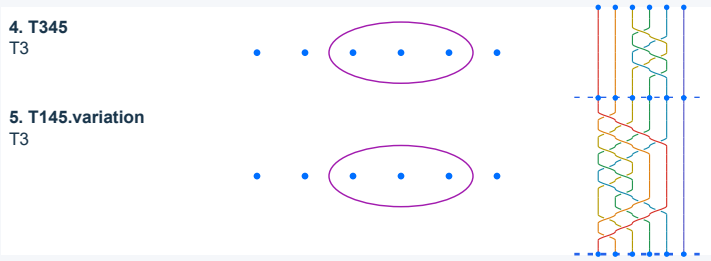
D - step 004

Combine the two matched T345.square

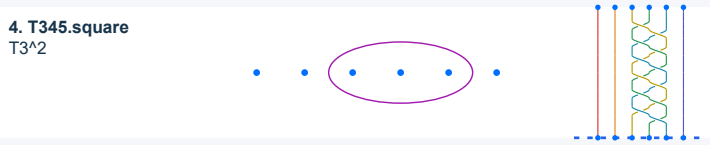
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

3 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



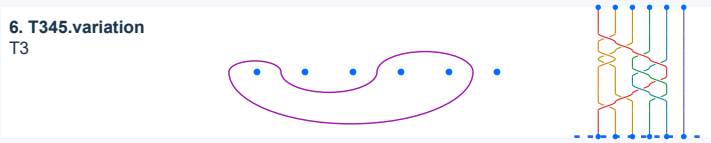
D - step 005

PC half lantern on the actual 145 root; expose its positive correction

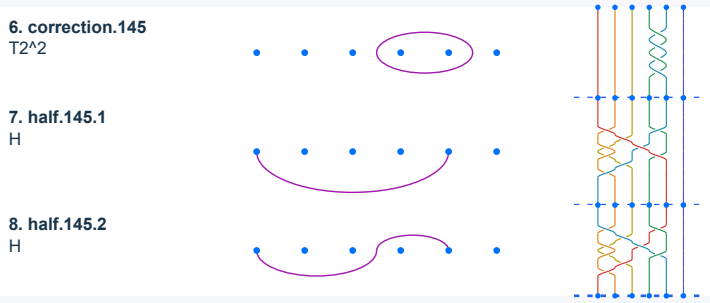
DISK SUBSTITUTION. Every ordered actual input/output and the full named local proof are checked. No sphere redraw or deletion is included.

5 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



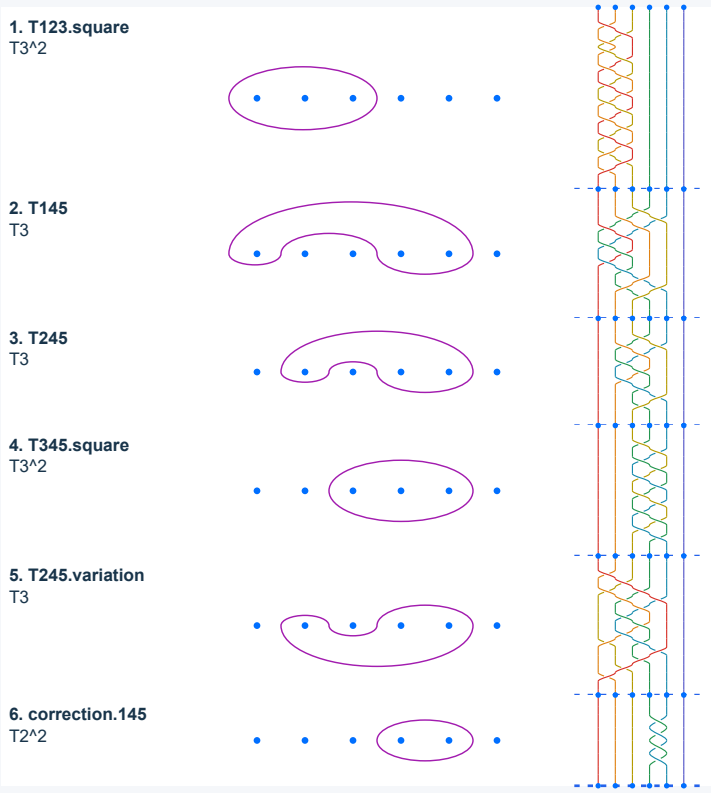
D - step 006

Move ONLY the new positive correction to the explicit LEFT ledger

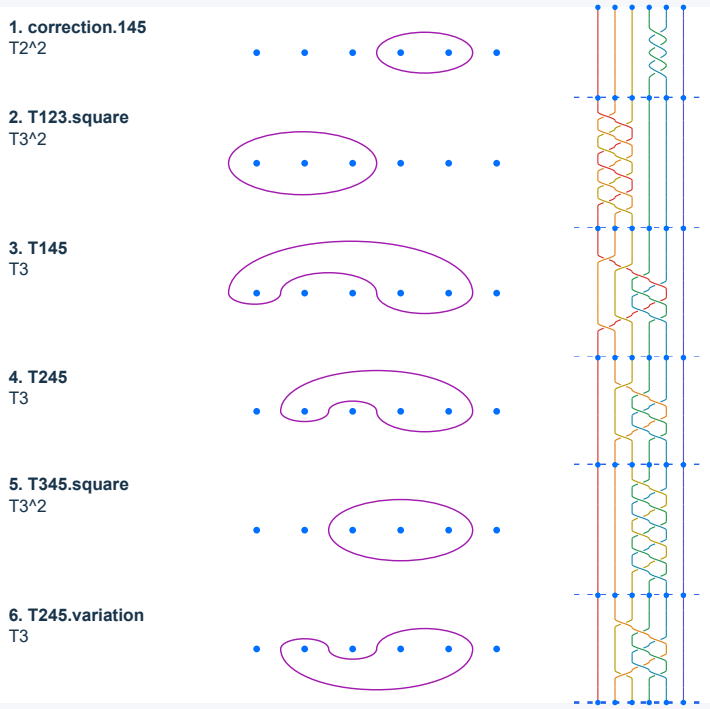
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

0 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



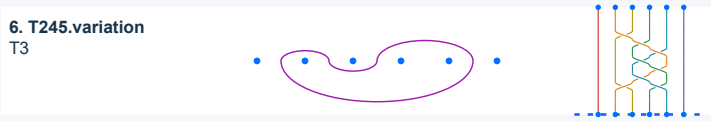
D - step 007

PC half lantern on the actual 245 root; expose its positive correction

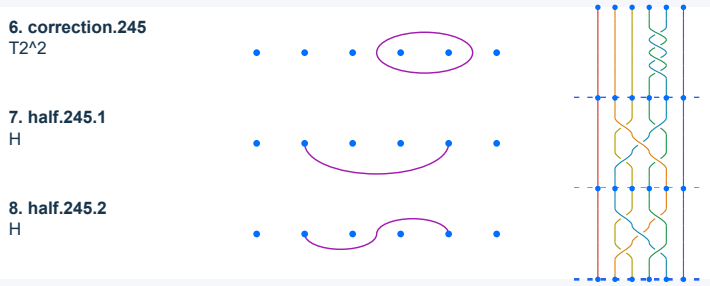
DISK SUBSTITUTION. Every ordered actual input/output and the full named local proof are checked. No sphere redraw or deletion is included.

5 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



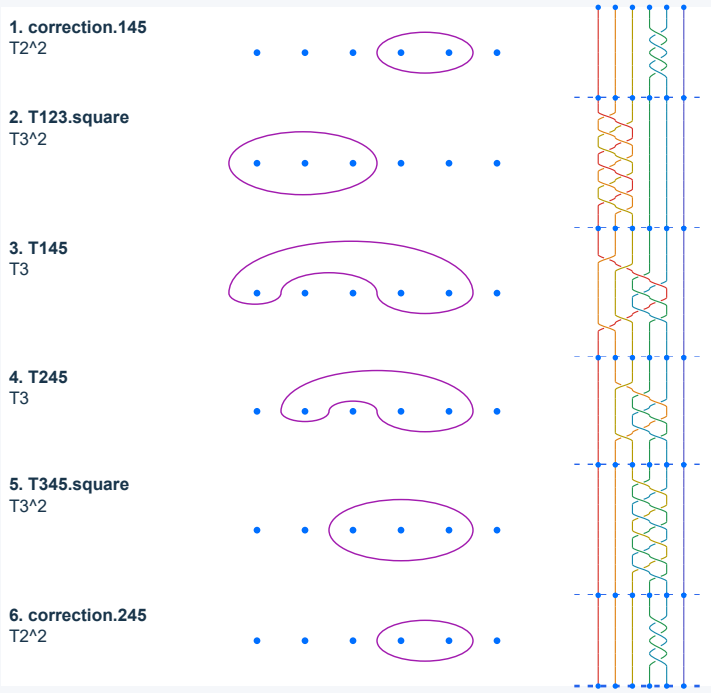
D - step 008

Move ONLY the new positive correction to the explicit LEFT ledger

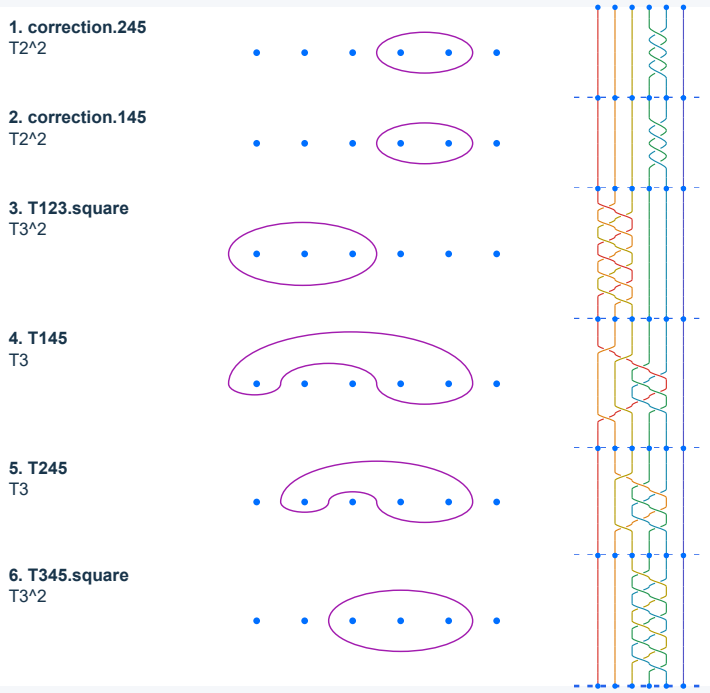
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



D - step 009

Combine the two matched positive.compensation

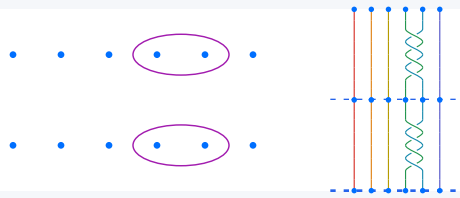
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

0 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before

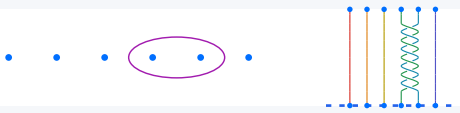
1. correction.245
T2^2

2. correction.145
T2^2



After

1. positive.compensation
T2^4



D - step 010

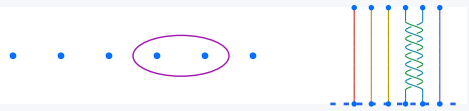
Cancel the SAME recorded Q45^4 prefix from BOTH sides of the relation

LEFT LEDGER CANCELLATION. Cancel the SAME explicit positive Q45^4 prefix from BOTH sides of the proved relation. This is not a sphere cap or deletion from a fixed-product fibration.

0 unchanged displayed factors above; 8 below. Only the changed interval is shown. The equal positive prefix is canceled from BOTH SIDES of the proved relation. It is not removed from a fixed-product fibration.

Before

1. positive.compensation
T2^4



After

Identity / no factors in this interval.

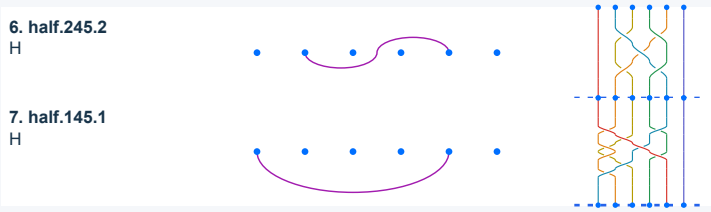
D - step 011

Move half7 up to6

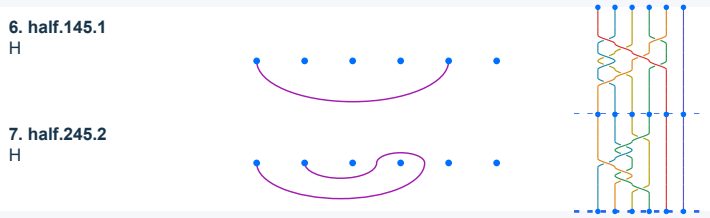
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

5 unchanged displayed factors above; 1 below. Only the changed interval is shown.

Before



After



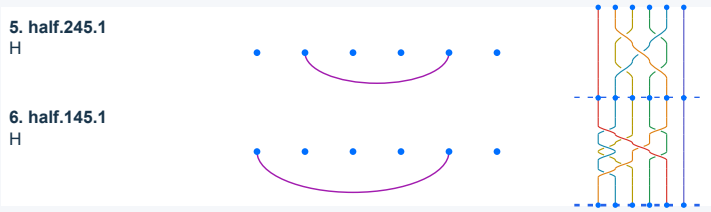
D - step 012

Move half5 down to6

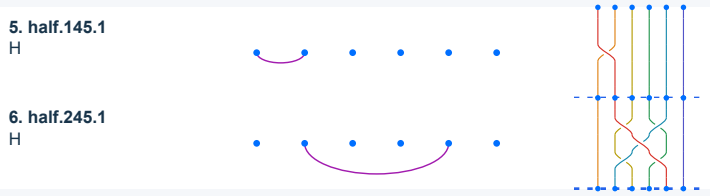
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

4 unchanged displayed factors above; 2 below. Only the changed interval is shown.

Before



After



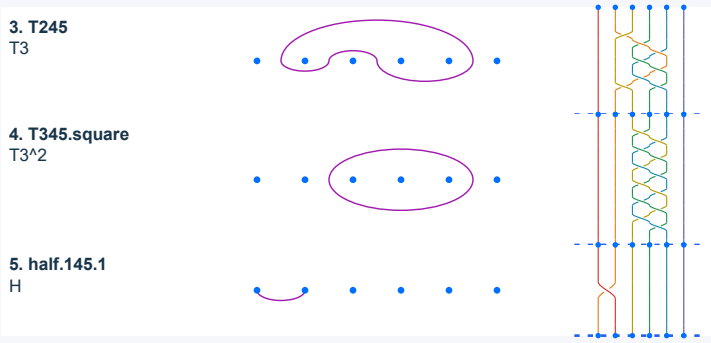
D - step 013

Move half5 up to3, with both adjacent crossings retained

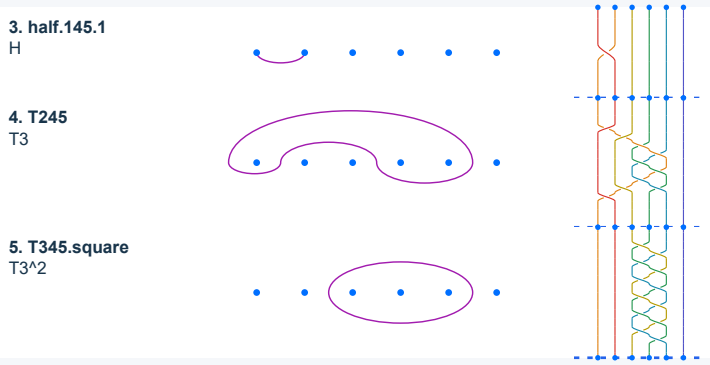
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

2 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



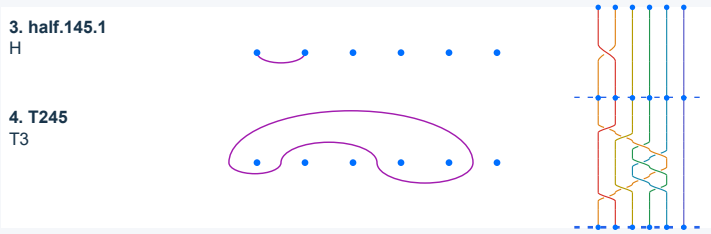
D - step 014

Move root4 up to3; the two ordinary triple roots now match

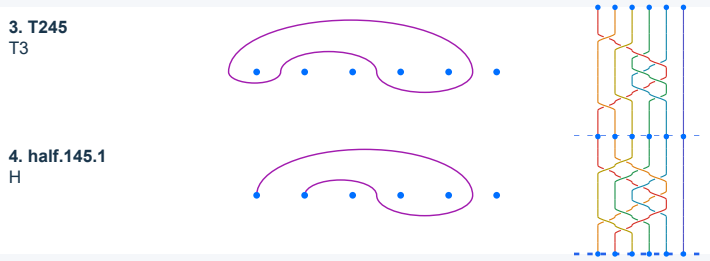
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

2 unchanged displayed factors above; 4 below. Only the changed interval is shown.

Before



After



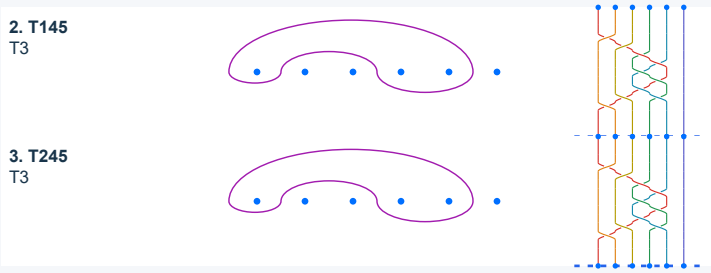
D - step 015

Combine the two matched third.separating.square

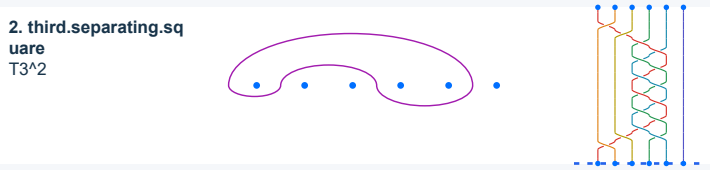
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

1 unchanged displayed factors above; 5 below. Only the changed interval is shown.

Before



After



Complete final type (4,3) factorization

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



Closure certificates and precise scope

The final product has the exact marked-sphere identity action and the separately checked chosen genus-two lift +I. This is not a claim of equality to the trivial disk braid, an unquotiented spherical-braid equality, a framed section lift, or an unproved identification with a particular literature curve chart.

Final chosen lift matrix: [[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1]]

Source-pinned full certificate: 4-3.json; graphical viewer: 4-3.html. Imported source manifest: ../fibration-sources/planar-mcg-2026-10-30/provenance.json. All sources, exact actual input/output matches, canonical PC daisies/half lanterns and parallel-copy corrections remain available with the repository.